

A Lean-Verified 51/50 Cutoff for Four-Arc Minimizers

The Cañete–Miranda–Vittone strip-density problem

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ABSTRACT

For the planar strip density equal to 1 on $\{|y| \leq 1\}$ and to $\lambda > 1$ outside, Cañete, Miranda Jr., and Vittone reduced the unresolved part of their classification to the possible optimality of a symmetric four-arc candidate. We give a machine-checked exclusion of every such candidate for $\lambda \geq 51/50$. The formal theorem is universal in the candidate curvature, includes the limiting curvature $h = 1$, and quantifies minimality against genuine measurable coordinate regions with weighted area and complete-frontier weighted perimeter. On the exact interval $[51/50, 9/7]$, Lean constructs an equal-area regular type-(iii) competitor with strictly smaller perimeter. Above $9/7$, the proof splices to the CMV two-cap-to-one-cap replacement, realized as an admissible three-arc carrier with the same area and smaller perimeter. The local transcendental work is discharged by 1,139 exact-rational cells whose seams and union are checked by Lean. The final theorem and its source-coordinate corollaries compile in Lean 4 using only the standard logical axioms reported by `#print axioms`.

Main result. Let $\lambda \geq 51/50$. Every modeled CMV type-(iv) four-arc candidate satisfying the density jump and Snell incidence law has an admissible equal-area competitor of smaller weighted perimeter. Consequently, it is not a weighted-perimeter minimizer. For a canonical regular source profile, the only remaining possible density range is

$$1 < \lambda < \frac{51}{50}.$$

Lean declaration: `CMVModeledCutoff.candidate_not_isWeightedPerimeterMinimizer_from_51_50`.

1. Statement, scope, and novelty

Consider the lower-semicontinuous planar density

$$f_\lambda(x, y) = \begin{cases} 1, & |y| \leq 1, \\ \lambda, & |y| > 1, \end{cases} \quad \lambda > 1.$$

For a sufficiently regular carrier $E \subset \mathbb{R}^2$, the associated weighted area and perimeter are

$$A_\lambda(E) = \int_E f_\lambda \, d\mathcal{L}^2, \quad P_\lambda(E) = \int_{\partial E} f_\lambda \, d\mathcal{H}^1.$$

The source problem is formulated for finite-perimeter sets and reduced boundaries. Cañete, Miranda Jr., and Vittone (CMV) prove existence, connectedness, symmetrization, and a geometric classification of candidates (Cañete et al. 2010, Proposition 3.6, Lemma 3.8, Proposition 3.9). Their Conjecture 3.12 predicts that type-(iv) candidates are never minimizing. Their Theorem 3.16 proves this for $\lambda \geq 4/\pi$ by replacing two exterior caps with one; Remark 3.17 explicitly says that $4/\pi$ is not optimal.

This paper proves the following formal cutoff.

Theorem 1 (Lean-verified modeled cutoff). If $\lambda \geq 51/50$, `candidate : FourArcCandidate  $\lambda$` , and `candidate.SatisfiesCMVTypeIVHypotheses`, then

$\neg$ `candidate.IsWeightedPerimeterMinimizer`.

The exact Lean statement is:

```
theorem candidate_not_isWeightedPerimeterMinimizer_from_51_50
  {lam : ℝ}
  (hlower : (51 / 50 : ℝ) ≤ lam)
  (candidate : FourArcCandidate lam)
  (hcandidate : candidate.SatisfiesCMVTypeIVHypotheses) :
  ¬ candidate.IsWeightedPerimeterMinimizer
```

It appears verbatim in `CMVModeledCutoff.lean`, lines 21–38. The proof is not a numerical sample and not a theorem about one selected curvature: the conclusion holds for every candidate in the formal family.

A logically equivalent corollary is:

```
theorem type_four_minimizer_implies_lambda_lt_51_50 ... :
  lam < (51 / 50 : ℝ)
```

and, after using the recorded density jump $1 < \lambda$:

```
theorem type_four_minimizer_open_range_51_50 ... :
  1 < lam ∧ lam < (51 / 50 : ℝ)
```

Both are in `CMVModeledCutoff.lean`, lines 40–61.

1.1 What “type-(iii) is better” means here

The comparison has two rigorous branches.

1. **Exact source-family type-(iii) comparison.** For every $\lambda \in [51/50, 9/7]$ and every admissible type-(iv) curvature, Lean produces a regular type-(iii) assembly with exactly equal weighted area and strictly smaller weighted perimeter.
2. **Three-arc cap replacement.** For $\lambda > 9/7$, Lean uses the CMV two-cap-to-one-cap construction. Its carrier has the type-(iii) topology: two strip-side circular arcs, one exterior circular arc, and the exposed density-one interface segment. It is a genuine equal-area admissible competitor and has smaller perimeter. It is not asserted to satisfy the stationary equal-curvature equation of the canonical type-(iii) candidate family.

This distinction matters. The full formal claim is the exclusion of type (iv) for every $\lambda \geq 51/50$. The stronger literal statement that a *stationary, equal-curvature* type-(iii) candidate beats every type-(iv) candidate is machine-checked on $[51/50, 9/7]$; above that interval the checked competitor is the cap replacement. Calling the latter stationary would add a premise not present in the Lean theorem.

2. From the CMV paper to formal parameters

CMV identify four candidate families. Their type (iii) has one exterior cap and two arcs through the strip; type (iv) has reflected exterior caps above and below the strip together with two strip-side arcs. Equations (26) and (27) give area and

perimeter in terms of generalized curvature h and principal angles (Cañete et al. 2010, 18–19). CMV record $0 < h \leq 1$ for type (iv), and their numerical plots suggest that type (iii) always beats type (iv) (Cañete et al. 2010, fig. 6 and the paragraph following equation (27)).

The formalization begins from coordinate geometry rather than postulating the scalar formulas.

2.1 Four-arc candidates

`FourArcCandidate.lean` defines the geometric data

$$0 < h \leq 1, \quad 0 < \alpha < \frac{\pi}{2},$$

then derives the common radius, strip core, two reflected caps, carrier, weighted area, and weighted perimeter. The source-specific predicate is only

```
structure SatisfiesCMVTypeIVHypotheses : Prop where
  density_jump : 1 < lam
  snell_incidence : lam * cos candidate.alpha = candidate.h
```

See `FourArcCandidate.lean`, lines 75–93. In particular, this predicate does **not** assume an area formula, a perimeter formula, a competitor, an inequality, or nonminimality. The principal-branch identity $\alpha = \arccos(h/\lambda)$ is proved from the incidence law and angle bounds (`alpha.eq_arccos`). The CMV type-(iv) formulas are then derived from the constructed boundary pieces (`candidate_perimeter_formula` and `candidate_area_formula`).

Minimality is not restricted to another scalar record. It quantifies over every formal `AdmissibleCompetitor` of the same weighted area:

```
def IsWeightedPerimeterMinimizer : Prop :=
  ∀ competitor : AdmissibleCompetitor lam,
    competitor.WeightedArea = candidate.WeightedArea →
      candidate.WeightedPerimeter ≤ competitor.WeightedPerimeter
```

Thus one concrete equal-area region with smaller perimeter refutes minimality.

2.2 Genuine type-(iii) carriers

`TypeThreeAssembly.lean` implements the regular CMV type-(iii) family on $0 < h_3 < 1$. It defines the closed coordinate carrier, its major/semicircular/minor exterior-cap branches, the two strip-side arc traces, and the possibly degenerate lower interface segment. The common radius is $R = 1/h_3$; branch selection is explicit rather than hidden in a numerical inverse function. The module proves measurability, finite-perimeter realization, weighted-area identities, and complete-frontier weighted-perimeter identities.

The analytic-to-geometric bridge is `FourArcCandidate.not_isWeightedPerimeterMinimizer_of_analytic_typeThree`: if the scalar type-(iii) area equals the type-(iv) area and its perimeter is strictly smaller, Lean constructs the `TypeThreeAssembly`, obtains its finite-perimeter realization, and contradicts candidate minimality.

2.3 Complete-frontier perimeter

The earlier scalar argument could not by itself certify that a formula was the perimeter of the displayed region. The current development closes that gap in `FrontierPerimeter.lean`. It proves exact frontier decompositions, zero $\mathcal{H}^1$ mass of finite join sets, pairwise almost-disjointness of arc measures, and equality between explicit boundary sums and the weighted integral over the complete topological frontier. The same module treats four-arc, type-(iii), and replacement assemblies.

These theorems use Hausdorff measure on the Euclidean realization, not a Manhattan proxy and not a sum of disconnected lengths. Endpoints are included set-theoretically and shown to have zero one-dimensional Hausdorff measure.

3. The scalar comparison on [51/50, 9/7]

The coordinate geometry reduces the stationary comparison to exact scalar identities. The notation below is the one used by the retained certificate:

$$h_4 = \cos x, \quad q_3 = \cos w, \quad h_3 = \frac{1 + \cos w}{2}.$$

For type (iv), set

$$B_4 = \lambda y + \frac{\pi}{2} - x, \quad d_4 = \sin y - \sin x, \quad N_4 = B_4 + \cos x d_4.$$

For type (iii), set

$$B_3 = \lambda v + \pi - w, \quad d_3 = \sin v - \sin w, \quad N_3 = B_3 + (\cos w + 2)d_3.$$

Four equations determine the stationary type-(iv) fold and its selected equal-area type-(iii) comparison:

$$\begin{aligned} C_4 &= \cos x - \lambda \cos y, \\ F &= \cos x (\sin y - \sin x) - \sin x \sin y (\lambda y + \frac{\pi}{2} - x), \\ C_3 &= \cos w - \lambda \cos v, \\ E &= 2 \cos^2 x N_3 - (1 + \cos w)^2 N_4. \end{aligned}$$

Here $C_4 = F = 0$ is the incidence/stationarity system for type (iv), while $C_3 = E = 0$ is the incidence/equal-area system for type (iii). The sign-transfer quantity is

$$G = \cos x (B_3 + d_3) - (1 + \cos w) B_4.$$

Under the proved domain and denominator guards, G has the sign of $P_3 - P_4$. Each retained cell proves $G < 0$.

The formulas and their derivatives are formalized in `LeanSuffixAnalytic.lean`. `SameCurvatureArea.lean` and `EqualAreaEnvelope.lean` turn one certified stationary comparison into an all-curvature statement. The decisive theorem `allCurvature_typeThreeImprovement` returns, for every $0 < h_4 \leq 1$, a regular h_3 with equal area and smaller perimeter.

At the candidate layer, `CompactCellAssembly.Conditions.candidate_not_isWeightedPerimeterMinimizer` packages the certified stationary pair, proves the descending type-(iii) branch condition, invokes the all-curvature envelope, and realizes the resulting competitor geometrically. This is why the final theorem covers arbitrary candidate curvature and the endpoint $h_4 = 1$ rather than only a stationary sample.

4. Exact-rational coverage

The compact numerical work is proof-producing. Floating-point root solvers are used to propose boxes, but Lean accepts only rational endpoints and exact inequalities. Transcendental values are enclosed using rational Taylor bounds; interval ordering, face signs, derivative signs, denominator positivity, and all seams are replayed by the kernel.

Component	Closed density interval	Cells	Formal role
Left prototype	[51/50, 102001/100000]	1	Starts the new cutoff and proves the first seam.
Middle-face tiling	[102001/100000, 25781/25000]	113	Covers the narrow source-to-compact transition.
Right prototype	[25781/25000, 33/32]	1	Reaches the compact certificate exactly.
Compact certificate	[33/32, 9/7]	1,024	Covers the remaining scalar suffix.
Total	[51/50, 9/7]	1,139	No rational seam gap.

The first three rows compose in `MiddleFaceCells.candidate_not_isWeightedPerimeterMinimizer_from_51_50`. The compact cells compose in `CompactCells1003To1023.bboxes0To1023_candidate_not_isWeightedPerimeterMinimizer`. The final cell ends at the exact rational 9/7, not a rounded decimal.

4.1 What a cell proves

A compact cell contains a rational interval in λ , a slab for the stationary type-(iv) root, and a slab for the equal-area type-(iii) root. Its `Conditions` proof establishes:

- strict physical domains for both curvatures;
- existence and uniqueness of the type-(iv) fold root;
- existence and selected-branch uniqueness of the type-(iii) equal-area root;
- strict negativity of the type-(iii) fold on the selected slab;
- strict negativity of $P_3 - P_4$;
- exact compatibility of adjacent density intervals.

For a representative generated leaf, see `CompactCells1003To1023.Box1003`. The leaf contains only rational literals and proofs built from the shared checker. The reusable checker lives in `CompactCellCertificate.lean`, and the candidate-facing composition is in `CompactCellAssembly.lean`.

The middle-face path uses the same principle with slanted four-dimensional bricks. Its reusable contract is `MiddleFaceAssembly.lean`, and its generated leaves are grouped by `MiddleFaceCells.lean`.

5. The high-density cap replacement

CMV's proof at $\lambda \geq 4/\pi$ replaces the two equal exterior caps of a type-(iv) candidate by one cap of twice the Euclidean area and exposes the opposite interface chord (Cañete et al. 2010, Theorem 3.16). If L is the old chord length, A one old cap area, and arc is the length of a circular arc with unit chord and prescribed enclosed area, strict improvement is equivalent to

$$\lambda L \text{arc}(2A/L^2) + L < 2\lambda L \text{arc}(A/L^2).$$

The formal development improves the numerical density threshold for this argument to the exact decimal rational

$$1.2581840884 = \frac{3145460221}{2500000000}.$$

`CapReplacement.lean` constructs the replacement carrier and proves that its upper cap has exactly twice the old Euclidean cap area. It covers minor, semicircular, and major branches. The exposed lower segment is proved to lie on the density-one interface and on the actual frontier (`exposed_lower_chord_on_frontier`).

`GeometricBridge.lean` proves exact area preservation and transfers the certified arc inequality to strict weighted-perimeter improvement. Finally, `CandidateExclusion.cmv_type_four_not_minimizing` instantiates minimality with that concrete region.

Since

$$1.2581840884 < \frac{9}{7},$$

the high-density result covers every $\lambda > 9/7$. The final composition is a three-way exact case split:

```
by_cases hface : lam ≤ (33 / 32 : ℝ)
-- 1, 113, and 1 face/prototype cells
...
by_cases hcompact : lam ≤ (9 / 7 : ℝ)
-- 1,024 compact cells
...
-- cap replacement, using 1.2581840884 < 9/7
```

This proof is visible at `CMVModeledCutoff.lean`, lines 29–38. There is no uncovered seam at 33/32 or 9/7.

6. Source-coordinate transfer

The modeled theorem is already a theorem about actual coordinate carriers and weighted integrals, but connecting an arbitrary source presentation to the canonical coordinates is a separate logical step.

`CMVSourceBridge.lean` defines `CanonicalTypeIVProfile`, its independent source-coordinate carrier, and the conversion `toCandidate`. It proves that the converted candidate satisfies the CMV incidence law and that every regular modeled candidate comes from such a profile. Horizontal translations preserve both weighted area and complete-frontier weighted perimeter (`HorizontallyCongruent` and `invariance` theorems).

For a generic external source semantics, the exact transfer theorem names its two assumptions rather than hiding them:

```
theorem sourceCarrier_not_isMinimizer_from_51_50
  (hlower : (51 / 50 : ℝ) ≤ lam)
  (source : SourcePerimeterSemantics lam)
  (sourceCarrier : Set PlanePoint)
  (profile : CanonicalTypeIVProfile lam)
  (normalization : source.NormalizationWitness sourceCarrier profile)
  (compatibility : source.CompatibleWithModel profile) :
  ¬ source.IsMinimizer sourceCarrier
```

See `CMVModeledCutoff.lean`, lines 63–83.

For the canonical complete-frontier source semantics implemented in the repository, those compatibility obligations are discharged internally. The result is:

```
theorem canonicalCarrier_minimizer_open_range_51_50
  (profile : CanonicalTypeIVProfile lam)
  (hmin : (canonicalFrontierSemantics lam).IsMinimizer profile.carrier) :
  1 < lam ∧ lam < (51 / 50 : ℝ)
```

See `CMVModeledCutoff.lean`, lines 103–118.

Scope boundary. CMV’s geometric classification and regularity arguments are cited, not re-formalized. The Lean theorem excludes every regular canonical source type-(iv) profile and every external source carrier satisfying the explicit normalization and reduced-boundary compatibility contract. It does not claim a new formalization of CMV’s existence or symmetrization theorems. The global conjecture remains open only on $1 < \lambda < 51/50$.

7. Trust and audit

The proof uses Lean 4 (Moura and Ullrich 2021) and mathlib (The mathlib Community 2020). The repository pins the compiler in `lean-toolchain` and pins dependencies in `lake-manifest.json`.

7.1 Trusted and untrusted components

Component	Role	Trusted for the theorem?
Lean kernel	Checks elaborated proof terms	Yes
mathlib definitions and theorems	Analysis, topology, measure theory, algebra	Yes, as imported theory
Rational certificate generators	Propose boxes and bounds	No
Floating-point root finding	Finds candidate witnesses	No
Generated <code>.lean</code> files	Supply rational propositions and proof terms	Only after kernel checking
Python mutation/replay scripts	Independent diagnostics	No
PDF/HTML rendering	Presentation only	No

The essential point is proof-producing computation: a bad floating-point guess can make generation fail, but it cannot make Lean accept a false rational inequality. Coverage is also a theorem-level composition, so omission of a cell creates a missing interval rather than silently passing a loop.

The focused publication audit lists the exported cutoff declarations in `CMVCutoffAssumptions.lean`. `#print axioms` reports only the standard logical axioms used throughout mathlib:

```
propext
Classical.choice
Quot.sound
```

No theorem in the cutoff chain depends on a project-local `axiom`, `sorry`, `admit`, or `native_decide` shortcut.

8. Reproduction

A clean checkout contains all Lean sources, generated exact certificates, build configuration, bibliography, manuscript source, stylesheet, and the paper build script.

8.1 Check the theorem

From the repository root:

```
cd projects/cmV-strip-density/proof
lake build CMVModeledCutoff
lake env lean CMVCutoffAssumptions.lean
```

The first command compiles the theorem and its complete dependency closure. The second prints the axiom dependencies of the public declarations. To build the entire formal development, including every generated cell:

```
cd projects/cmV-strip-density/proof
lake build
```

The relevant roots and pinned dependency are declared in `lakefile.toml`.

8.2 Rebuild the paper

The paper is generated from this Markdown source and the adjacent stylesheet:

```
cd projects/cmV-strip-density/reports
./build-lean-verified-cmv-cutoff.sh
```

The script invokes Pandoc with citation processing and then Chromium’s print engine. Its outputs are `lean-verified-cmv-cutoff.html` and `lean-verified-cmv-cutoff.pdf`. The manuscript’s direct source links target the same repository paths used by the Lean commands above.

9. Conclusion

The 2010 CMV analysis isolated type (iv) as the remaining obstruction and proved its exclusion for $\lambda \geq 4/\pi$. The present formal development lowers the verified cutoff to

$$\lambda \geq \frac{51}{50}.$$

The result is stronger than the earlier scalar certificate in three ways. First, it is a Lean theorem about every modeled type-(iv) candidate, not a standalone table of numerical signs. Second, the equal-area type-(iii) comparison on $[51/50, 9/7]$ is realized as a genuine finite-perimeter coordinate competitor whose weighted area and complete-frontier weighted perimeter agree with the scalar formulas. Third, the theorem transfers to independent source coordinates through explicit normalization and boundary-semantics contracts, with the canonical source carrier requiring no external bridge assumption.

The mathematical work left by this theorem is sharply delimited: decide the punctured near-one interval $1 < \lambda < 51/50$ and, separately, formalize as much of the global CMV existence/classification argument as desired. Neither task is silently included in the cutoff proved here.

References

- Cañete, Antonio, Michele Miranda Jr., and Davide Vittone. 2010. “Some Isoperimetric Problems in Planes with Density.” *The Journal of Geometric Analysis* 20: 243–90. <https://arxiv.org/abs/0906.1256>.
- Moura, Leonardo de, and Sebastian Ullrich. 2021. “The Lean 4 Theorem Prover and Programming Language.” *Automated Deduction—CADE 28*, Lecture notes in computer science, vol. 12699: 625–35. https://doi.org/10.1007/978-3-030-79876-5_37.
- The mathlib Community. 2020. “The Lean Mathematical Library.” *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs*, 367–81. <https://doi.org/10.1145/3372885.3373824>.